# Xiang Li 李想

PhD Student in Algebra & Number Theory, The University of Edinburgh

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## Research Interests

Arithmetic geometry and number theory, particularly nonabelian Chabauty methods, rational and integral points of algebraic curves over number fields, and $p$-adic periods.

## Education

**PhD in Algebra and Number Theory** Sep 2023 – 2027 (expected)
*School of Mathematics (Hodge Institute), The University of Edinburgh*
*Supervisor: Professor Minhyong Kim*

**MASt in Mathematics (Part III)** Oct 2022 – Jun 2023
*Department of Pure Mathematics and Mathematical Statistics, University of Cambridge*

**BSc (Hons) Mathematics, First Class Honours** Sep 2020 – Jul 2022
*School of Mathematics, The University of Edinburgh*

**BSc in Mathematics** Sep 2018 – Jun 2020
*School of Mathematics, South China University of Technology*
*2+2 joint programme with the University of Edinburgh; both degrees awarded in July 2022.*

## Preprints

Nonabelian Chabauty for the thrice-punctured line over cyclotomic fields, with Minhyong Kim and Martin Lüdtke. arXiv:2609.01128, September 2026.

Polylogarithmic Chabauty–Kim loci over number fields, with Martin Lüdtke. arXiv:2608.20615, August 2026.

## Awards and Scholarships

Carnegie PhD Scholarship, 2023 – 2027.

Napier Medal and Gangadhar Balwant Gadgil Prize for the best 4th year student, 2021/22.

Arthur Erdelyi Prize for distinguished performance in the Degree Examinations for year 3 Mathematics, 2020/21.

Dr Nigel Suess Prize for distinguished performance in 'Introduction to Number Theory', 2020/21.

College Vacation Scholarship, 2020/21.

National Scholarship (China), 2018/19.

## Talks

**Invited seminar talks**

Nonabelian Chabauty methods over number fields. Glasgow Algebra and Number Theory Seminar, University of Glasgow, UK. 14 October 2026.

Rigid geometry and Coleman function. The GEARS seminar, University of Glasgow, UK. 7 February 2025.

Euler's totient theorem and the prime number theorem (with Ella Yu). London Learning Lean seminar, Imperial College London. 16 February 2023.

**Contributed talks**

Polylogarithmic Chabauty–Kim loci over number fields. Young Researchers in Algebraic Number Theory VIII, University of Bristol, UK. 3 September 2026.

Chabauty–Kim methods over number fields: $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Speed talk, ChaBONNty Conference, Max Planck Institute for Mathematics, Bonn, Germany. 2 July 2026.

$S$-integral points on the thrice-punctured line over cyclotomic fields. Young Researchers in Algebraic Number Theory VII, University of Nottingham, UK. 5 September 2025.

Introduction to Chabauty–Kim methods on $S$-unit equations. Young Researchers in Algebraic Number Theory VI, University of Oxford, UK. 2 August 2024.

Tannakian category, unipotent completion, and de Rham fundamental group. Examples Showcase, GlaMS, ICMS, Edinburgh, UK. 1 December 2023.

## Teaching

**Tutor, School of Mathematics, The University of Edinburgh**
Responsibilities include leading small-group tutorials and problem-solving sessions, and marking coursework.

2026–27, Semester 1: Honours Differential Equations

2025–26, Semester 2: Fundamentals of Pure Mathematics

2025–26, Semester 1: Introduction to Mathematics at University

2024–25, Semester 2: Proofs and Problem Solving; Fundamentals of Pure Mathematics

2024–25, Semester 1: Introduction to Linear Algebra

2023–24, Semester 2: Calculus and its Applications; Algebraic Topology

**MathsBase Tutor, University of Edinburgh** 2024–25, Semester 1
Drop-in mathematics support for first- and second-year undergraduates.

**Teaching training:** School of Mathematics Teaching Stream, October 2024.

## Software

PolylogNF — Sage code accompanying the preprints *Polylogarithmic Chabauty–Kim loci over number fields* and *Nonabelian Chabauty for the thrice-punctured line over cyclotomic fields.*

prime-number-theorem — a collaborative project towards formalising the prime number theorem in Lean, with Huajian Xin, Ella Yu and others.

## Selected Conferences and Workshops

2026 — ChaBONNty Conference, Max Planck Institute for Mathematics, Bonn

2024 — Arithmetic, Geometry, Space and Time, ICMS, Edinburgh

2024 — The Mordell Conjecture 100 Years Later, MIT

2024 — Gauge Fields in Arithmetic, Topology and Physics, ICMS, Edinburgh

2024 — Winter Workshop Chabauty–Kim, Heidelberg University

Last updated: October 2026